Measuring AI Patent Enforceability with Large Language Models: Evidence from Recentive

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Research Question

- Does the Recentive decision lead banks to reprice and restructure syndicated loan contracts for firms with greater pre-existing exposure to vulnerable applied ML patents?

- Do banks respond more strongly when vulnerable AI patents are more likely to matter as collateral, such as for firms with prior patent pledging activity, low tangible assets, or near-term refinancing needs?

Institutional Setting

- The eligiblility of patent claim

- 35 U.S.C. Section 101 defines patent eligible subject matter as *any new and useful process, machine, manufacture, or composition of matter.*

Alice framework
(Alice Corp. v. CLS Bank Int'l, 2014)

Step 1: Is the claim directed to a patent-ineligible abstract idea?
Step 2: Is there an inventive concept that transforms the claim into eligible subject matter?

within-AI legal boundary →

Recentive boundary
(Recentive. In Aon Re, Inc. v. Zesty.AI, Inc., 2025)

The decision holds vulnerable claims ineligible when they do no more than apply generic ML to a new data environment, without improving the ML model itself.

Literature Review

- **Patents as collaterals**

- Patents support debt when lenders can identify, value, enforce, and transfer the underlying rights (Mann, 2018; Hochberg et al., 2018). Patent invalidation reduces leverage, suggesting that legal strength is priced in debt markets (Horsch et al., 2021).

- **IP protection decline**

- When IP protection is weakened, firms adjust protection strategies, including greater secrecy and NDA use (Acikalin et al., 2022).

- **AI and economic consequences**

- Previous studies mainly study growth, labor, innovation, and equity valuation rather than bank loan contract design (Babina et al., 2024; Kogan et al., 2023; Eisfeldt & Schubert, 2025). This study adds legal enforceability and pledgeability of AI knowledge as a financing channel.

Hypothesis Developments

- **H1 (Price Channel)**. Following Recentive, firms with higher pre-period Recentive Vulnerability Scores face higher all-in drawn spreads on newly originated syndicated loans.

- **H2 (Collateral-Package and Control-Rights Reconfiguration)**. Higher exposure to Recentive- vulnerable claims leads to tighter non-price loan terms, including more stringent collateral requirements, stronger IP-specific reporting, negative pledges, IP-transfer restrictions, shorter maturities, smaller facilities, and tighter covenants.

- **H3 (Asset-Form Substitution)**. Firms with greater exposure to vulnerable applied-ML patent claims are expected to exhibit partial substitution from patent-form protection toward secrecy-form protection.

Data for Main Analysis

The empirical dataset links AI patent text, firm financials, disclosures, and syndicated loan contracts.

AI patent data

PatentsView utility patents
USPTO AIPD 2023
CPC G06N and keyword supplements

→

Firm linkages

Patent assignment records
KPSS patent–PERMNO
WRDS CCM to GVKEY

→

Loan contracts

DealScan facilities
2018Q1–2026Q1
SEC Exhibit 10 credit agreements

→

Financials & disclosure

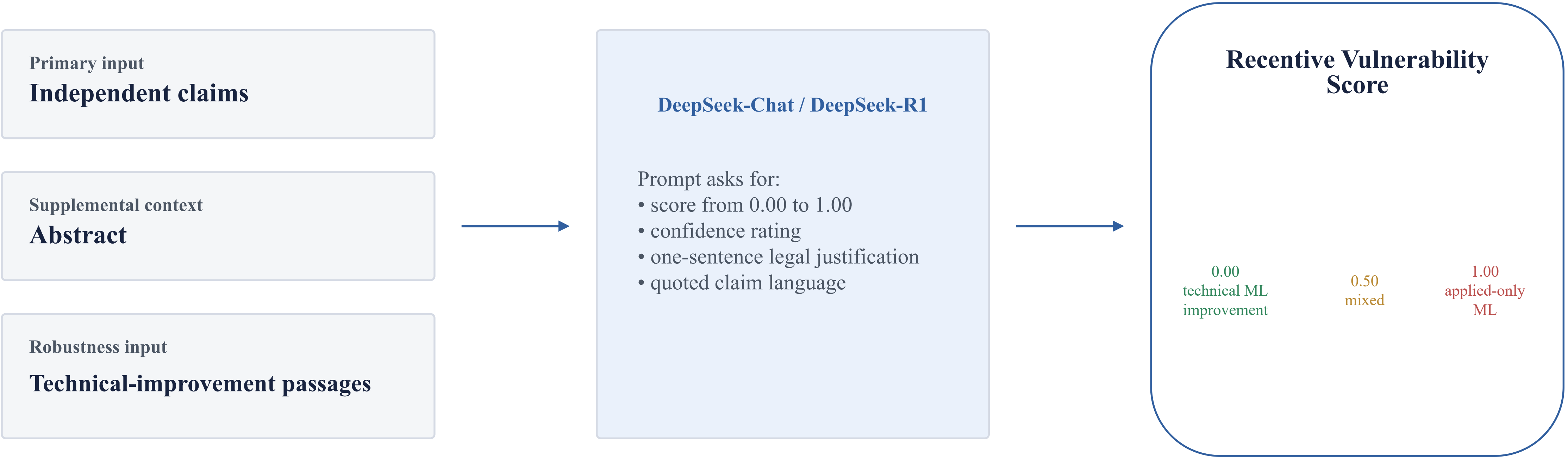
Compustat controls
CRSP event-study returns
10-K secrecy markers via LLM

Primary sample U.S. public AI-patenting firms with at least three AI patents granted before 2025Q1 and at least one DealScan facility in the sample window.

Identification focuses within the AI-patenting universe; non-AI firms serve as external placebo observations.

Measuring Vulnerability Through AI

- After collection of patent information, vulnerability score is constructed through the textual analysis of AI



- Human validation: a stratified sample of 300 AI patent claims is hand-coded on the same 0–1 vulnerability scale to benchmark the LLM classifier.

Measuring Vulnerability Through AI

- Prompt

“ You are a patent attorney applying the Federal Circuit's two-step Alice test as articulated in *Recentive Analytics, Inc. v. Fox Corp*. Read the patent material below and output a score from 0.00 to 1.00 reflecting the degree to which this claim resembles Recentive-vulnerable applied-only ML claim language. Anchors: 0.00 = the claim recites a clear technological improvement to the machine learning model itself (a new architecture, loss function, training procedure, or representation); 0.50 = mixed or ambiguous; 1.00 = the claim recites only the application of established machine learning techniques to a new data environment, business problem, or domain, without disclosing a technological improvement to the model. Provide the score, a confidence rating, and a one-sentence justification quoting the specific claim language that drove the decision.”

Measuring Vulnerability Through AI

$$\text{Vulnerability}_{i,2025Q1} = \frac{\sum_{p \in P_i^{AI}} Score_p}{\left| P_i^{AI} \right|}$$

P_i^{AI} = firm i's stock of AI patents granted before 2025Q1.

Alternative measurements as robustness tests

Citation-weighted mean

weights measured before Apr. 18, 2025

Top-quartile exposure

high-treatment indicator

Predicted vulnerable stock

count-based exposure

Vulnerable stock / assets

scaled by firm balance sheet

Baseline Estimation Model

$$y_{f,i,t} = \beta(\text{Vulnerability}_{\text{i,pre}} \times \text{Post}_{\text{t}}) + \Gamma X_{i,t-1} + \alpha_i + \delta_{j(f),t} + \mu_{k(i),t} + \varepsilon_{f,i,t}$$

where f indexes facility, i indexes borrower, t indexes loan-origination quarter, *j(f)* denotes lead arranger, and *k(i)* denotes Fama-French industry. $Post_t$ equals one if the loan origination date is on or after April 18, 2025. The specification includes borrower fixed effects α_{i}, lead-arranger by quarter fixed effects $\delta_{\text{j}(f),t}$, and industry by quarter fixed effects $\mu_{k(i),t}$. Borrower fixed effects absorb time-invariant differences across firms. Lead-arranger by quarter fixed effects control for time-varying credit supply conditions at the lender level. Industry by quarter fixed effects absorb sectoral shocks, including contemporaneous changes in AI adoption or AI sentiment. Standard errors are clustered by borrower and quarter.

Heterogeneity tests

$$y_{f,i,t} = \beta_1(\text{Vulnerability}_{\text{i,pre}} \times Post_t \times Z_i) + \text{Lower Order Terms} + \alpha_{\text{i}} + \delta_{j(f),t} + \mu_{k(i),t} + \varepsilon_{f,i,t}$$

Z_i denotes a predetermined heterogeneity variable measured before the *Recentive* decision. The main variables are pre-period patent-pledger status, low tangibility, near-term refinancing need, and borrowing from an IP-specialised lender. Each indicator is fixed at the pre-period level and is therefore not contaminated by post-*Recentive* contract responses.

Subsample test

$$y_{p,f} = \beta(\text{Vulnerability}_{\text{i(p),pre}} \times Post_f) + \alpha_p + \eta_{c(f)} + \varepsilon_{p,f}$$

p denotes a borrower lead-arranger pair. α_{p} absorbs all time-invariant characteristics of the borrower lender relationship. This within-borrower-within-lender approach extends the identification strategy pioneered by Khwaja and Mian (2008), which exploits variation across loan transactions of the same borrower with different lenders to isolate supply-side effects, by additionally exploiting time variation around the *Recentive* decision within the same pair. The estimating sample consists of pairs with at least one loan in the pre-period from 2023 Q1 to 2025 Q1 and at least one loan in the post-period from 2025 Q3 to 2026 Q1.

Robustness tests

Event-study diagnostics

Estimate quarterly leads and lags from 2022 Q1 to 2026 Q1.

Key diagnostic: no systematic pre-trends before the April 18, 2025 decision.

Special attention to 2024 Q4–2025 Q1 because oral argument occurred on October 1, 2024, creating possible anticipation.

Donut specifications

Exclude loans originated within 30 days of the decision date.

Repeat using a 60-day exclusion window.

Purpose: avoid classifying loans negotiated before Recentive but signed shortly after as post-shock responses.

Conservative post-period

Redefine Post = 1 only for loans originated in 2025 Q3 or later.

This allows time for banks to update legal-risk assessments and renegotiate contract terms.

Results should remain directionally consistent under this stricter timing definition.

Robustness tests

Within-AI placebo

Replace Vulnerability with core-machine-learning share.

Prediction: null effect.

Interpretation: Recentive should affect applied-only ML claims more than patents claiming genuine model improvements.

Non-AI and non-U.S. placebos

Use non-AI software-patent share × Post.

Use European or Chinese AI-patent share × Post.

Prediction: null effects, helping rule out general software shocks and global AI sentiment shocks.

Fake event dates

Re-estimate using placebo event dates in 2023 Q2 and 2024 Q2.

Prediction: no treatment effect around these artificial dates.

Purpose: test whether results are driven by pre-existing trends.

IV robustness

Instrument firm vulnerability using leave-one-out examiner leniency toward generic/applied-only ML claims within art unit × year.

Reported as robustness, not primary identification, because the exclusion restriction is potentially contestable.

Preliminary Results

Table A1. Recentive Vulnerability Score Applied to Litigated *Recentive* Patents

Patent No.	Grant Year	AIPD ML Flag	AIPD ML Score	Vulnerability Score	Confidence
10,911,811	2021	1	0.878	0.85	High
10,958,957	2021	0	0.123	0.85	High
11,386,367	2022	0	0.272	0.95	High
11,537,960	2022	0	0.418	0.95	High
Mean				0.9	

Preliminary Results

Table A2. Inter-Model Robustness: DeepSeek-Reasoner vs DeepSeek-Chat

Statistic	DeepSeek-Reasoner	DeepSeek-Chat
Mean	0.747	0.773
Standard deviation	0.347	0.231
25th percentile	0.85	0.85
50th percentile	0.95	0.85
75th percentile	0.95	0.85
Maximum	1	0.95
Valid scores	4,970	4,970

Preliminary Results

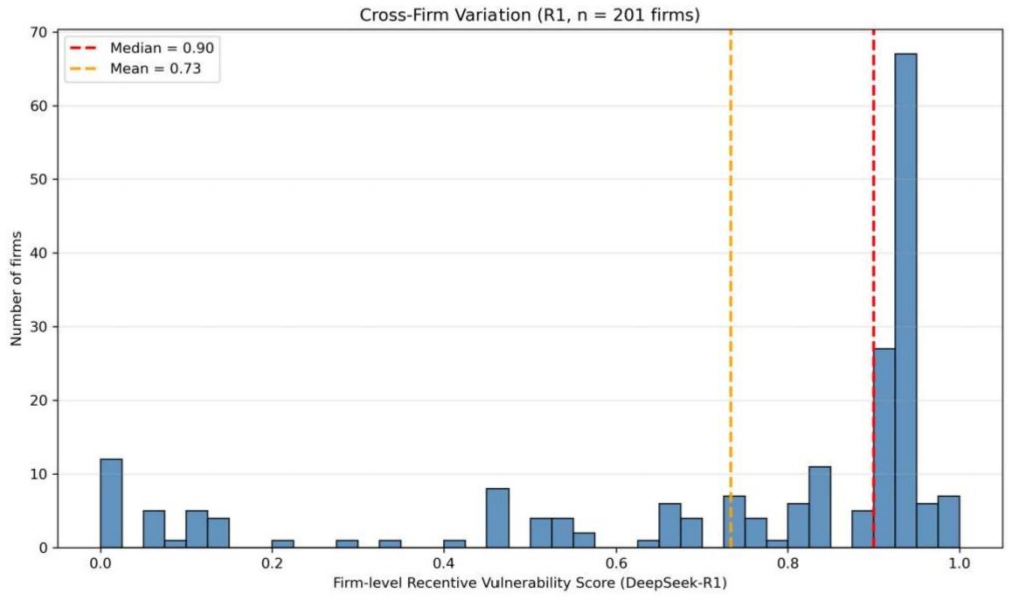


Figure A1. Cross-Firm Distribution of Firm-Level Recentive Vulnerability

Preliminary Results

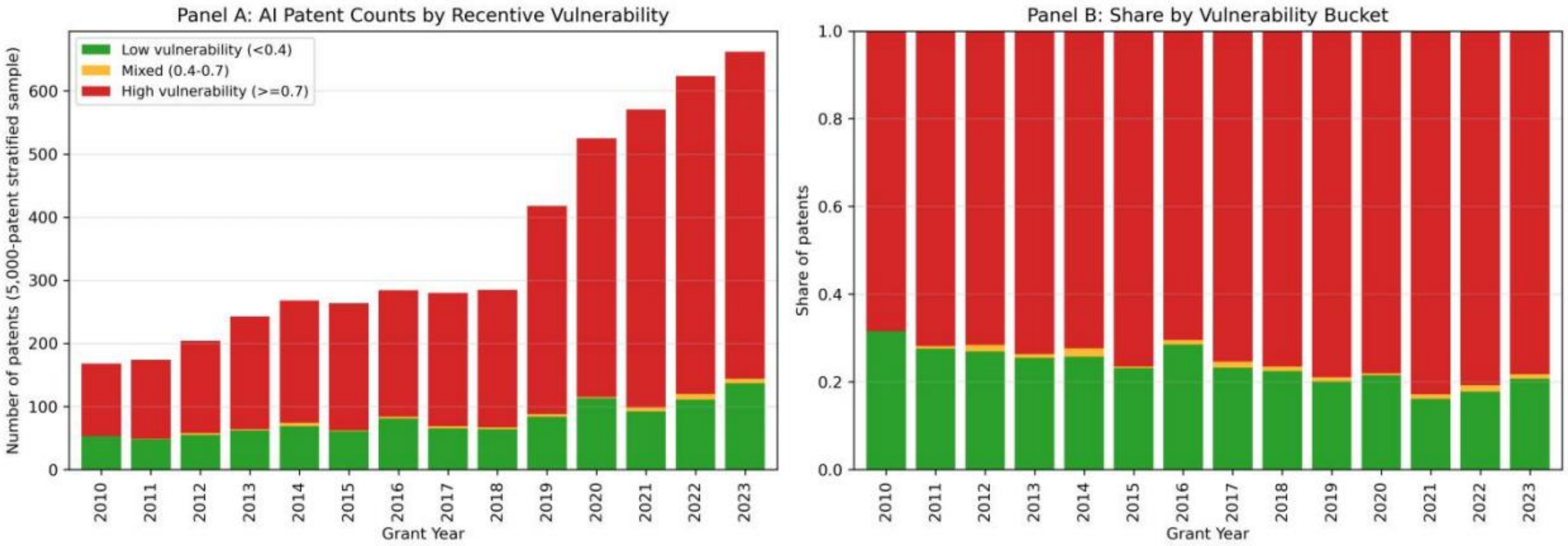


Figure A2. Distribution of Recentive Vulnerability by Patent Grant Year

Contributions and Takeaways

Recentive turns AI patent enforceability into an observable financing variable.

1. **Within-AI legal shock**

   Uses the applied-ML versus model-improvement boundary rather than a broad software-patent shock.

2. **AI-based empirical measurement**

   Transforms legal doctrine into a scalable LLM-derived vulnerability score over patent claims.

3. **Debt contract architecture**

   Studies spreads jointly with covenants, collateral, maturity, facility size, and monitoring rights.

4. **Asset-form channel**

   Shows how AI knowledge may move from patent-form collateral toward secrecy-form protection.

Main takeaway: AI affects corporate finance not only through productivity or valuation, but through the legal pledgeability of AI knowledge.